Physics: Motion

Course Information

Course Name: Physics – Motion

Level: Beginner / Intermediate

Prerequisites: Basic mathematics, including arithmetic, fractions, and simple algebra

Estimated Duration: 8–10 Hours

Course Description

Motion is one of the fundamental topics in physics. It describes how the position of an object changes with time. In this course, students will learn the basic concepts of motion, including position, distance, displacement, speed, velocity, acceleration, and time. Students will study different types of motion, motion graphs, equations of motion, free fall, and relative motion. The course also introduces practical applications of motion in everyday life.

Learning Outcomes

After completing this course, students will be able to:

- Define motion and rest.
- Understand position and reference points.
- Distinguish between distance and displacement.
- Calculate speed and average speed.
- Understand velocity and average velocity.
- Define acceleration.
- Calculate acceleration from changes in velocity.
- Distinguish between scalar and vector quantities.
- Understand uniform and non-uniform motion.
- Interpret distance-time graphs.
- Interpret velocity-time graphs.
- Apply equations of motion.
- Solve basic free-fall problems.
- Understand relative motion.
- Solve basic numerical problems involving motion.
- Apply concepts of motion to real-world situations.

Lesson 1: Introduction to Motion

Learning Objectives

After this lesson, students will be able to:

- Define motion.
- Define rest.
- Understand the importance of a reference point.
- Identify examples of motion in everyday life.

What is Motion?

An object is said to be in **motion** when its position changes with time relative to a reference point.

For example, a moving car changes its position relative to a road, building, or tree.

Examples of Motion

Examples include:

- A car moving on a road.
- A ball falling from a height.
- Earth moving around the Sun.
- A person walking.
- A train moving along a track.
- A fan rotating around its axis.

Lesson 2: Rest and Reference Point

Rest

An object is said to be at **rest** when its position does not change with time relative to a reference point.

For example, a book lying on a table is at rest relative to the table.

Reference Point

A **reference point** is a fixed point used to determine whether an object is moving or at rest.

Consider a passenger sitting inside a moving bus.

Relative to another passenger:

```text id="x8m4q2" Passenger → At rest

````
Relative to a person standing beside the road:

```text id="p6n9v3"
Passenger → In motion
````

Therefore, motion and rest depend on the reference point.

Lesson 3: Position

Position describes the location of an object relative to a reference point.

For example, suppose a car is 20 meters east of a tree.

Its position can be described as:

```text id="m7q2k8" 20 m east of the tree

````
Position is important because motion involves a change in position.

Lesson 4: Distance

Distance is the total length of the path traveled by an object.

Distance is a **scalar quantity**, meaning it has magnitude but no direction.

Example:

A student walks:

```text id="c4v8n1"
10 m forward
````

and then:

```text id="r6p2m9" 5 m backward

````
The total distance is:
````
````

````
```text id="x3k7q5"
Distance = 10 + 5
Distance = 15 m
````

SI Unit of Distance

The SI unit of distance is:

```text id="n8m4p2" meter (m)

````
Other units include:

- Kilometer (km)
- Centimeter (cm)
- Millimeter (mm)

Lesson 5: Displacement

Displacement is the change in position of an object from its initial
position to its final position.

Displacement is a **vector quantity**, meaning it has both magnitude and
direction.

Example:

A person walks:

```text id="q5v9k3"
10 m east
````

and then:

```text id="m2x7p8" 5 m west

````
The displacement is:

```text id="c6n1r4"
10 m east - 5 m west = 5 m east
````

Therefore:

```text id="j8q3m6" Displacement = 5 m east

````

Lesson 6: Distance vs Displacement

Distance	Displacement
Scalar quantity	Vector quantity
Has magnitude only	Has magnitude and direction
Depends on the actual path	Depends only on initial and final positions
Always positive or zero	Can be positive, negative, or zero depending on
direction
Can be greater than displacement	Magnitude is never greater than distance

---

Example

A runner completes one full lap of a circular track and returns to the
starting point.

Suppose the track length is:

```text id="p4m8x2"
400 m
````

Distance:

```text id="v7n3q9" 400 m

````
Displacement:

```text id="k2c6m5"
0 m
````

because the runner ends at the starting position.

Lesson 7: Scalars and Vectors

Scalar Quantity

A scalar quantity has magnitude only.
````

Examples:

- Distance
- Speed
- Time
- Mass
- Temperature

Vector Quantity

A vector quantity has magnitude and direction.

Examples:

- Displacement
- Velocity
- Acceleration
- Force

Lesson 8: Time

Time measures the duration of an event.

The SI unit of time is:

```text id="x9m3q7" second (s)

````
Other units include:

- Minute
- Hour
- Day

Common Conversions

```text id="n5v8k2"
1 minute = 60 seconds
````

```text id="p3m7x1" 1 hour = 60 minutes

````
Therefore:
````
````

````
```text id="q6c2r9"
1 hour = 3600 seconds
````

Lesson 9: Speed

Speed is the distance traveled per unit time.

The formula is:

```text id="m8q4v6" Speed = Distance / Time

````
or:

```text id="x2p9k5"
v = d/t
````

where:

- `v` = speed
- `d` = distance
- `t` = time

SI Unit of Speed

The SI unit is:

```text id="r7n3c1" meter per second (m/s)

````

Example

A car travels 100 meters in 5 seconds.

```text id="k4m8q2"
v = d/t
````

```text id="p6x1n9" v = 100/5

````
Therefore:
````

````
```text id="c3v7m5"
v = 20 m/s
````

Lesson 10: Average Speed

Average speed is the total distance traveled divided by the total time taken.

```text id="q8m2x6" Average Speed = Total Distance / Total Time

````

Example

A car travels:

```text id="n4p9k1"
100 km
````

in:

```text id="v6m3q8" 2 hours

````
Average speed:

```text id="x5c7r2"
Average Speed = 100/2
````

```text id="m9q4n6" = 50 km/h

````

Lesson 11: Velocity

Velocity is the rate of change of displacement with respect to time.

The formula is:

```text id="p2x8m5"
Velocity = Displacement / Time
````

or:

```text id="n7q3v1" v = Δx/Δt

````
Velocity is a **vector quantity**, so direction must be specified.

Example:

```text id="c6m9k4"
20 m/s east
````

is a velocity.

Lesson 12: Speed vs Velocity

Speed	Velocity
Scalar quantity	Vector quantity
Uses distance	Uses displacement
Has no direction	Has direction
Cannot be negative in ordinary treatment	Can have positive or negative values depending on chosen direction

---

Example

A car travels at:

```text id="x8q2m7" 60 km/h east

````
Its:

- Speed = 60 km/h
- Velocity = 60 km/h east

Lesson 13: Acceleration

Acceleration is the rate at which velocity changes with time.

The formula is:

```text id="m4v7n2"
a = Δv/Δt
````

or:
````

```text id="q9p3x6" a = (v-u)/t

````
where:

- `a` = acceleration
- `v` = final velocity
- `u` = initial velocity
- `t` = time

SI Unit

The SI unit of acceleration is:

```text id="k5m8c1"
m/s²
````

Example

A car increases its velocity from:

```text id="n3q7p2" 10 m/s

````
to:

```text id="x6m1v9"
30 m/s
````

in:

```text id="r8k4c5" 5 s

````
Acceleration:

```text id="m2q9n6"
a = (30 - 10)/5
````

```text id="p7v3x1" a = 20/5

````
```text id="c4m8q2"
a = 4 m/s²
````

Lesson 14: Positive and Negative Acceleration

Acceleration can be positive or negative depending on the chosen direction.

Positive acceleration generally means velocity is increasing in the positive direction.

Negative acceleration can indicate a decrease in velocity in a chosen positive direction.

Negative acceleration is often called **deceleration** when it results in slowing down.

Lesson 15: Uniform Motion

An object is in **uniform motion** when it covers equal distances in equal intervals of time.

Example:

A car travels:

```text id="v5n8m2" 10 m every second

```
Its speed remains constant.

Characteristics

In uniform motion:

- Speed remains constant.
- Equal distances are covered in equal time intervals.
- Acceleration is zero if velocity is constant.

Lesson 16: Non-Uniform Motion

An object is in **non-uniform motion** when it covers unequal distances in
equal intervals of time.

Examples:

- A car accelerating from a traffic light.
- A falling object.
- A cyclist changing speed.
- A bus stopping at a station.

```
```

````
# Lesson 17: Distance-Time Graphs

A distance-time graph shows how distance changes with time.

Usually:

- Time is placed on the horizontal axis.
- Distance is placed on the vertical axis.

---

## Constant Speed

A straight line with constant slope represents constant speed.

The slope of a distance-time graph represents speed:

```text id="x7m3q9"
Speed = Change in distance / Change in time
````

Stationary Object

A horizontal line means the object is not changing its position with time.

Therefore:

```text id="p4k8n2" Speed = 0

```

Lesson 18: Velocity-Time Graphs

A velocity-time graph shows how velocity changes with time.

Usually:

- Time is on the horizontal axis.
- Velocity is on the vertical axis.

Constant Velocity

A horizontal line indicates constant velocity.

```

````
## Constant Acceleration

A straight sloping line indicates constant acceleration.

The slope of a velocity-time graph gives acceleration:

```text id="m6q1v8"
Acceleration = Change in velocity / Change in time
````

Area Under a Velocity-Time Graph

The area under a velocity-time graph represents displacement.

For a simple rectangular region:

```text id="n8x5p3" Displacement = velocity × time

````

Lesson 19: Equations of Motion

For motion with constant acceleration, three important equations are used.

First Equation

```text id="q2m7k9"
v = u + at
````

where:

- `u` = initial velocity
- `v` = final velocity
- `a` = acceleration
- `t` = time

Second Equation

```text id="x4p8n1" s = ut + 1/2 $at^2$

````
where:

- `s` = displacement
- `u` = initial velocity
- `a` = acceleration
````

````
- `t` = time

---

## Third Equation

```text id="m9v3q6"
v² = u² + 2as
````

Lesson 20: Using Equations of Motion

Example

A car starts from rest and accelerates at:

```text id="k5x2m8" 4 $m/s^2$

````
for:

```text id="p7n1q4"
5 s
````

Find the final velocity.

Given:

```text id="c3v9m6" $u = 0$ $a = 4\ m/s^2$ $t = 5$ s

````
Use:

```text id="x8q2p5"
v = u + at
````

Therefore:

```text id="n6m4k1" $v = 0 + (4)(5)$

````
```text id="r7v3q9"
v = 20 m/s
````

````

Lesson 21: Finding Displacement

Using the same example:

```text id="m2x8c6" u = 0 a = 4 m/s² t = 5 s

```
Use:

```text id="p9q4n7"
s = ut + 1/2at²
```

Substitute:

```text id="v6m1k3" s = (0)(5) + 1/2(4)(5²)

```
```text id="x8r2q5"
s = 2 × 25
```

Therefore:

```text id="n4m7c9" s = 50 m

```

Lesson 22: Motion Under Gravity

Objects falling toward Earth experience acceleration due to gravity.

Near Earth's surface:

```text id="k3p8v1"
g ≈ 9.8 m/s²
```

For basic calculations, `g` may sometimes be approximated as:

```text id="q6m2x9" 10 m/s²

```

Free Fall

An object is in free fall when gravity is the only significant force acting on it.
```
```

````
Examples:

- A ball dropped from a height.
- An object falling from a building.
- A stone falling toward Earth.

Air resistance may be ignored in idealized problems.

---

# Lesson 23: Object Dropped from Rest

Suppose an object is dropped from rest.

Then:

```text id="m8q3n5"
u = 0
````

Using:

```text id="x4p7k2" v = u + gt

````
we get:

```text id="n6m9c1"
v = gt
````

Example

An object falls for 3 seconds.

Using:

```text id="r2v8m4" g = 9.8 m/s$^2$

````
Final velocity:

```text id="p5q1x7"
v = 9.8 × 3
````

```text id="c9m3n6" v = 29.4 m/s

````
directed downward.

Lesson 24: Distance Fallen Under Gravity

For an object dropped from rest:

```text id="k7x2m8"
s = 1/2gt²
````

Example:

An object falls for 2 seconds.

```text id="n4p9v1" g = 9.8 $m/s^2$

````
Therefore:

```text id="q6m3c8"
s = 1/2 × 9.8 × 2²
````

```text id="x5r7n2" s = 4.9 × 4

````
```text id="m8k1p6"
s = 19.6 m
````

Lesson 25: Motion in a Straight Line

Motion along a straight path is called **linear motion** or one-dimensional motion.

Examples:

- A train moving along a straight track.
- A car traveling on a straight road.
- An object falling vertically.

In one-dimensional motion, we can choose a positive direction and a negative direction.

For example:

```text id="v3q8m5" Right = Positive Left = Negative

````

Lesson 26: Circular Motion

Circular motion occurs when an object moves along a circular path.

Examples:

- A satellite orbiting Earth.
- A stone attached to a string and rotated.
- The tip of a fan blade.
- A car moving around a circular track.

Even if the speed is constant, the velocity changes because the direction
continuously changes.

Therefore, circular motion involves acceleration.

Lesson 27: Uniform Circular Motion

In **uniform circular motion**, an object moves around a circular path at
constant speed.

Although speed remains constant, velocity changes because direction changes.

The acceleration is directed toward the center of the circle and is called
centripetal acceleration.

The formula is:

```text id="q9m4x2"
a_c = v²/r
````

where:

- `a_c` = centripetal acceleration
- `v` = speed
- `r` = radius

Lesson 28: Relative Motion

Motion can appear different to different observers.

Relative motion describes the motion of one object as observed from another object or reference frame.
````

Example:

A passenger sitting in a moving train is:

- At rest relative to another passenger.
- Moving relative to a person standing beside the track.

Relative Velocity

For objects moving in the same direction:

```text id="m6p2v8" v_relative = $v_1$ - $v_2$

~~~~
For objects moving in opposite directions:

```text id="x4n7q1"
v_relative = v₁ + v₂
~~~~

---

# Lesson 29: Average Velocity

Average velocity is:

```text id="p8m3k6" Average Velocity = Total Displacement / Total Time

~~~~
Example:

An object has a displacement of:

```text id="q2v9n4"
100 m east
~~~~

in:

```text id="m7x1c5" 20 s

~~~~
Therefore:

```text id="r6p8k2"
Average Velocity = 100/20
~~~~

```text id="n3m5q9" = 5 m/s east
```

```
---

# Lesson 30: Motion in Everyday Life

The concepts of motion are used in many areas.

## Transportation

Engineers calculate speed, acceleration, braking distance, and travel time.

## Sports

Athletes and coaches analyze:

- Speed
- Velocity
- Acceleration
- Projectile motion
- Reaction time

## Engineering

Engineers study motion when designing:

- Vehicles
- Machines
- Elevators
- Robots
- Aircraft

## Space Science

Scientists study the motion of:

- Planets
- Moons
- Satellites
- Spacecraft

---

# Solved Problems

## Problem 1: Calculate Speed

A cyclist travels 200 meters in 20 seconds.

Find the speed.

### Solution
```

````
```text id="k8m3q5"
Speed = Distance / Time
````

```text id="x2v7n1" Speed = 200/20

````
Therefore:

```text id="p9c4m6"
Speed = 10 m/s
````

Problem 2: Calculate Acceleration

A car increases its velocity from 5 m/s to 25 m/s in 4 seconds.

Find acceleration.

Solution

```text id="m6q2x8" a = (v-u)/t

````
```text id="n4p7k1"
a = (25-5)/4
````

```text id="v8m3c5" a = 20/4

````
Therefore:

```text id="r2q9x6"
a = 5 m/s²
````

Problem 3: Calculate Final Velocity

An object starts from rest and accelerates at 3 m/s^2 for 6 seconds.

Find the final velocity.

Solution

Given:

```text id="k5m1v8" u = 0 a = 3 $m/s^2$ t = 6 s

````
Use:

```text id="q7x3n9"
v = u + at
````

```text id="p4m8c2" v = 0 + (3)(6)

````
Therefore:

```text id="n6q1r5"
v = 18 m/s
````

Problem 4: Calculate Displacement

A car starts from rest and accelerates at 2 m/s^2 for 10 seconds.

Find its displacement.

Solution

```text id="x8m4q7" u = 0 a = 2 $m/s^2$ t = 10 s

````
Use:

```text id="v3n9p2"
s = ut + 1/2at²
````

```text id="m6k1x8" s = 0 + 1/2(2)($10^2$)

````
```text id="q4p7c5"
s = 100 m
````

Problem 5: Free Fall

An object is dropped from rest for 2 seconds. Find its final velocity.

Take:

```text id="n8m2v6" g = 9.8 $m/s^2$

````
Solution

```text id="p5q9x1"
v = u + gt
````

Since:

```text id="c7m3k8" u = 0

````
we get:

```text id="r4v6n2"
v = 9.8 × 2
````

Therefore:

```text id="x1m8q5" v = 19.6 m/s

````
downward.

Practice Exercises

Exercise 1

A car travels 150 meters in 10 seconds.

Find its average speed.

Exercise 2

A runner covers 400 meters in 50 seconds.

Find the average speed.

Exercise 3

A vehicle increases its velocity from 10 m/s to 30 m/s in 5 seconds.

Find its acceleration.

````

````
## Exercise 4

A car starts from rest and accelerates at 5 m/s² for 4 seconds.

Find its final velocity.

---

## Exercise 5

A car starts from rest and accelerates at 3 m/s² for 6 seconds.

Find its displacement.

---

## Exercise 6

An object travels 80 meters east and then 30 meters west.

Find:

1. Total distance.
2. Displacement.

---

## Exercise 7

An object is dropped from rest and falls for 4 seconds.

Take:

```text id="k6m2p9"
g = 9.8 m/s²
````

Find:

1. Final velocity.
2. Distance fallen.

Exercise 8

A train travels at a constant velocity of 20 m/s for 30 seconds.

Find its displacement.

````

Exercise 9

Two cars travel in the same direction at:

```text id="x7n3q8" 30 m/s

````
and:

```text id="p5m9k2"
20 m/s
````

Find their relative velocity.

Exercise 10

An object moves in a circular path with a speed of 10 m/s and radius of 5 m.

Find its centripetal acceleration.

Use:

`text id="v4q8m1"`
`a_c = v²/r`

Mini Project

Analyze the Motion of a Moving Vehicle

Students will analyze the motion of a vehicle over a short period.

Record the vehicle's position at regular time intervals.

For example:

Time (s)	Position (m)
0	0
2	10
4	25
6	45
8	70

Students should:

1. Calculate the distance traveled.
2. Calculate average speed.
3. Calculate average velocity if direction is specified.
4. Determine whether the motion is uniform or non-uniform.
5. Create a distance-time graph.
6. Analyze the slope of the graph.
7. Describe how the vehicle's motion changes over time.

Final Quiz

1. What is motion?
2. What is rest?
3. What is a reference point?
4. What is position?
5. Define distance.
6. Define displacement.
7. What is the difference between distance and displacement?
8. What is a scalar quantity?
9. What is a vector quantity?
10. Give two examples of scalar quantities.
11. Give two examples of vector quantities.
12. What is speed?
13. What is the SI unit of speed?
14. What is average speed?
15. What is velocity?
16. What is the difference between speed and velocity?
17. What is acceleration?
18. What is the SI unit of acceleration?
19. What is uniform motion?
20. What is non-uniform motion?
21. What does the slope of a distance-time graph represent?
22. What does the slope of a velocity-time graph represent?
23. What does the area under a velocity-time graph represent?
24. State the first equation of motion.
25. State the second equation of motion.
26. State the third equation of motion.
27. What is free fall?
28. What is the approximate acceleration due to gravity near Earth's surface?
29. What is circular motion?
30. What is centripetal acceleration?
31. Why does an object in uniform circular motion have acceleration even when its speed is constant?
32. What is relative motion?
33. What is average velocity?
34. Give two real-world applications of motion.
35. Why is a reference point important when describing motion?

Course Summary

After completing this course, students should understand the fundamental concepts of motion and be able to solve basic numerical and conceptual problems involving moving objects.

The major concepts covered are:

- Motion
- Rest
- Reference points
- Position
- Distance
- Displacement
- Scalars
- Vectors
- Time
- Speed
- Average speed
- Velocity
- Average velocity
- Acceleration
- Positive and negative acceleration
- Uniform motion
- Non-uniform motion
- Distance-time graphs
- Velocity-time graphs
- Equations of motion
- Free fall
- Motion under gravity
- Linear motion
- Circular motion
- Centripetal acceleration
- Relative motion
- Real-world applications of motion

Next Step

After completing **Physics – Motion**, students can continue with **Force and Laws of Motion**, followed by **Work, Energy and Power**, **Gravitation**, **Waves**, and **Electricity**.